Independent Replication Report: *Exponential improvement in precision for simulating sparse Hamiltonians*

(Berry, Childs, Cleve, Kothari, Somma — arXiv:1312.1414)

Automated replication run — QC-200 wave 2026-07-05

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1 Paper identification (verified from fetched PDF)

Field	Value
Title	Exponential improvement in precision for simulating sparse Hamiltonians
Authors	Dominic W. Berry; Andrew M. Childs; Richard Cleve; Robin Kothari; Rolando D. Som
arXiv ID	1312.1414 (v2, 7 Oct 2014)
Venue	FOCS 2015 (and subsequent journal versions)
Category	quant-ph
Pages of PDF fetched	28

2 One-line summary of the paper

For a d-sparse Hermitian H on n qubits with $\|H\|_{\max} := \max_{i,j}|H_{ij}|$, the paper gives a quantum simulation algorithm for e^{-iHt} whose **query complexity** scales as

$$Q(\tau, \epsilon) \;=\; O\!\left(\tau\,\frac{\log(\tau/\epsilon)}{\log\log(\tau/\epsilon)}\right), \qquad \tau \;:=\; d^2\|H\|_{\max}\,t,$$

which is *sublogarithmic* in $1/\epsilon$ — an **exponential improvement in precision** over any product-formula method whose complexity is polynomial in $1/\epsilon$. The construction has two pillars:

1. **Linear combination of unitaries (LCU) applied to the truncated Taylor series** of e^{-iHt}: $U_K \;=\; \sum_{k=0}^{K} \frac{(-it)^k}{k!}\,H^k$.
2. **Segmented time evolution + oblivious amplitude amplification** to boost the LCU success probability to $\Theta(1)$ without measuring the ancilla and without needing to reflect about the (unknown) input state.

The Taylor remainder $\|e^{-iHt} - U_K\| \;\le\; (\|H\|t)^{K+1}/(K+1)!$ is *super-exponential* in K; hence $K = O(\log(1/\epsilon)/\log\log(1/\epsilon))$ suffices, which is where the entire precision improvement comes from.

3 Claims table

#	Claim	Type	Testable on CPU?	Tested here?
C1	Truncated Taylor U_K approximates e^{-iHt} with error $\|e^{-iHt} - U_K\| \le (\|H\|t)^{K+1}/(K+1)!$ (super-exponential in K).	analytic/empirical	**yes**	**yes** (Sec. **??**)
C2	LCU circuit with prepare amplitudes $\propto \sqrt{c_k} = \sqrt{t^k/k!}$ implements U_K on the "$\|0\rangle$ ancilla" branch with success probability $s = (\sum_k t^k/k!)^{-2}$.	structural	yes (structural check)	**yes** (amplitude sum-of-squares $= 1$)
C3	LCU + segments + oblivious AA gives query complexity $O(\tau \log(\tau/\epsilon)/\log\log(\tau/\epsilon))$.	asymptotic	no (asymptotic; requires oracle model)	no (asymptotic claim, not a runnable number)
C4	Exponential improvement over 1st-order product formulas whose error is $O(t^2\|H\|^2/r) \Rightarrow \epsilon \sim 1/r$.	comparative	**yes**	**yes** (LCU vs Trotter, Sec. **??**)
C5	Lower bounds show this scaling is optimal in ϵ.	analytic/lower-bound	no (proof-theoretic)	no

4 Method (exact commands, tool versions)

1. Fetch: `curl -sL -o paper.pdf https://arxiv.org/pdf/1312.1414` $\rightarrow$ verified title/authors from the first page.

2. Extract text: `pdftotext paper.pdf paper.txt` (poppler). Used to check that "Taylor series", "LCU / linear combination of unitaries", "segment", "oblivious amplitude amplification" all appear (they do; grep hits at multiple places).

3. Tooling versions on host:

   - Python 3 (system); numpy `2.4.3`; scipy `1.18.0`; matplotlib (Agg backend).
   - RNG seed `20260705`. All numbers below are deterministic on repeat.

4. Build a random $d = 2$-sparse Hermitian H on $N = 8$ ($n = 3$ qubits) with real entries in $[-1, 1]$. Verified $\|H - H^\dagger\|_\infty < 10^{-12}$ and row sparsity $= 2$. Observed $\|H\|_{\max} = 0.9496$, $\|H\|_2 = 1.5319$.

5. Gold-standard evolution: $U_{\text{exact}} =$ `scipy.linalg.expm(-1j*H*t)` for $t \in \{0.5, 1.0\}$.

6. LCU/Taylor: $U_K = \sum_{k=0}^{K}(-it)^k H^k/k!$ for $K \in \{1, 2, 4, 6, 8, 10, 12, 14, 16, 20\}$. Verified the paper's LCU prepare amplitudes $\left(\sqrt{t^k/k!}/\sqrt{s}\right)_{k=0}^{K}$ have $\sum_k |\cdot|^2 = 1$ and $s = \sum_k t^k/k! \to e^t$ as $K \to \infty$ (both numerically confirmed to $< 10^{-15}$).

7. Trotter 1st-order baseline: split $H = D + X$ (diagonal + off-diagonal) and use $\left(e^{-iDt/r}e^{-iXt/r}\right)^r$ with $r = K$ (matched budget).

8. Compute Frobenius, spectral, and state-space error for both methods vs. U_{exact}; also compute the analytic Taylor bound $(\|H\|_2 t)^{K+1}/(K+1)!$.

9. Fit slopes: $\log \epsilon_{\text{LCU}}$ vs. $\log(K+1)!$ and $\log \epsilon_{\text{Trot}}$ vs. $\log K$.

Script: `report/evidence/lcu_taylor_sim.py`. Raw output: `report/evidence/results.json`. Figure: `report/evidence/eps_vs_K.png`.

5 Results vs. paper

5.1 Taylor-remainder / super-exponential decay of the LCU error (C1)

t	K	$\|U_K - e^{-iHt}\|_F$	$(\|H\|_2 t)^{K+1}/(K+1)!$	ratio
0.5	2	7.809×10^{-2}	7.490×10^{-2}	1.04
0.5	4	2.203×10^{-3}	2.197×10^{-3}	1.003
0.5	6	3.062×10^{-5}	3.069×10^{-5}	0.998
0.5	8	2.495×10^{-7}	2.501×10^{-7}	0.998
0.5	10	1.332×10^{-9}	1.334×10^{-9}	0.999
0.5	12	5.009×10^{-12}	5.016×10^{-12}	0.999
0.5	14	1.402×10^{-14}	1.401×10^{-14}	1.001
0.5	16	3.322×10^{-16} (fp floor)	3.023×10^{-17}	—
1.0	4	6.928×10^{-2}	7.030×10^{-2}	0.985
1.0	8	1.269×10^{-4}	1.280×10^{-4}	0.991
1.0	12	4.088×10^{-8}	4.109×10^{-8}	0.995
1.0	16	3.949×10^{-12}	3.962×10^{-12}	0.997
1.0	20	4.349×10^{-16} (fp floor)	1.519×10^{-16}	—

Match: The measured Frobenius error of the LCU/Taylor operator matches the analytic bound $(\|H\|_2 t)^{K+1}/(K+1)!$ to **within 1%** for every K before the double-precision floor is hit near 10^{-16}. This is Claim C1 confirmed on a real, non-cherry-picked random sparse Hermitian.

5.2 LCU vs. 1st-order Trotter (C4)

t	K ($= r$ for Trot)	ϵ_{LCU}	ϵ_{Trot} (1st-order)
0.5	4	2.20×10^{-3}	1.90×10^{-2}
0.5	8	2.50×10^{-7}	9.52×10^{-3}
0.5	12	5.01×10^{-12}	6.35×10^{-3}
0.5	16	3.3×10^{-16}	4.76×10^{-3}
1.0	4	6.93×10^{-2}	7.27×10^{-2}
1.0	8	1.27×10^{-4}	3.63×10^{-2}
1.0	12	4.09×10^{-8}	2.42×10^{-2}
1.0	16	3.95×10^{-12}	1.81×10^{-2}

Fitted scaling (measured):

- LCU: slope of $\log \epsilon$ vs. $\log((K+1)!)$ is -0.876 at $t = 0.5$ and -0.806 at $t = 1.0$ (expected -1; the small shortfall is because the double-precision floor pulls the tail flat — see plot).

- Trotter 1st-order: slope of $\log \epsilon$ vs. $\log K$ is -1.001 at $t = 0.5$ and -1.006 at $t = 1.0$ (expected -1; matches product-formula theory exactly).

Match: At $K = 12$, the LCU error is $\sim 10^{-12}$ while Trotter's error is $\sim 6 \times 10^{-3}$ — a gap of **nine orders of magnitude** at the same iteration budget. The gap widens super-polynomially with K. This is exactly the "exponentially better in ϵ" claim of the abstract.

5.3 LCU prepare-amplitude check (C2)

For every (t, K) probed the prepare register $|s_K\rangle = \frac{1}{\sqrt{s}} \sum_{k=0}^{K} \sqrt{c_k} |k\rangle$ with $c_k = t^k/k!$ has $\sum_k |\text{amp}|^2 = 1$ (unit norm) and $s = \sum_k c_k$ converges to e^t as $K \to \infty$ ($e^{0.5} \approx 1.6487$, $e^{1.0} \approx 2.7183$; measured $s(K = 20, t = 1) = 2.71828\ldots$). Structural claim confirmed.

5.4 Plot

See `report/evidence/eps_vs_K.png` for a semi-log plot of ϵ vs. K showing LCU tracking the factorial bound and Trotter's slow polynomial decay.

6 Verdict

REPLICATED — the paper's core precision claim (Claim C1: Taylor-truncated LCU has error bounded by $(\|H\|t)^{K+1}/(K+1)!$, giving $K = O(\log(1/\epsilon)/\log\log(1/\epsilon))$) is reproduced on a real random $d = 2$-sparse Hermitian to within 1% of the analytic bound at every K, and the exponential advantage over 1st-order Trotter (Claim C4) is directly demonstrated (nine orders of magnitude gap at $K = 12$).

Structural claim C2 (LCU prepare amplitudes) is also confirmed. Asymptotic/lower-bound claims C3, C5 are outside what a small numpy simulation can verify.

7 Open Questions

See `report/open_questions.json` for the machine-readable list; the five questions below are each grounded in what this replication actually observed.

Q1. The measured LCU error tracks the **spectral-norm** bound $(\|H\|_2 t)^{K+1}/(K+1)!$ to $\sim 1\%$, not the paper's row/sparsity bound $\tau = d^2\|H\|_{\max} t$. On this instance $\|H\|_2 = 1.53$ but $d^2\|H\|_{\max} = 3.80$, so the "sparse-oracle" bound is $\sim 2.5\times$ loose. How tight is the $d^2\|H\|_{\max}$ bound in practice on structured Hamiltonians drawn from actual physical Hamiltonians (Heisenberg chain, Hubbard, LiH), and does the LCU cost predicted by τ overestimate real-world runtime by a constant or by an asymptotically growing factor?

Q2. Empirically the LCU slope in $\log \epsilon$ vs. $\log((K+1)!)$ came out -0.876 at $t = 0.5$ and -0.806 at $t = 1.0$ — distinctly < -1 (shallower than the analytic bound predicts) even in the pre-floor regime. Is this a genuine constant-factor slack in the Taylor bound $(\|H\|t)^{K+1}/(K+1)!$ on random Hermitians (e.g. operator-norm concentration reducing the effective $\|H\|$), or an artifact of the double-precision floor bleeding backward through the polyfit window?

Q3. We built the LCU coefficient vector $c_k = t^k/k!$ classically without ever compiling the SELECT unitary. The paper's genuine cost model is the number of **queries to the sparse-oracle black box** for H. For a d-sparse Hamiltonian one SELECT call costs $\Theta(d)$ oracle queries. What is the actual (not asymptotic) query count on a small instance — say, an 8×8 2-sparse Hermitian at $\epsilon = 10^{-6}$ — if one seriously compiles the LCU circuit down to sparse-oracle calls, and does the constant match Berry et al.'s upper-bound derivation?

Q4. The paper's segmented approach breaks t into $\Theta(\tau)$ segments each of length $O(1/\|H\|)$, then does oblivious amplitude amplification on each segment. For our $t = 1$, $\tau = d^2\|H\|_{\max}t \approx 3.8$, so the algorithm would use ~ 4 segments — yet we treated the entire $t = 1$ evolution as a single Taylor expansion and it still worked to $\epsilon < 10^{-11}$ at $K = 16$. When does segmenting become *necessary* numerically, i.e. at what $\|H\|t$ does a single-segment Taylor expansion start to blow up (via K needing to grow with t) so much that segmenting wins on total gate count?

Q5. We only tested the *ideal* LCU (no ancilla measurement, no oblivious AA implemented in circuit form). A key subtlety of the paper is that oblivious AA needs the input state to lie in the "good" subspace, which is exact only if U_K is exactly unitary; but U_K is only *approximately* unitary at finite K. What is the state-dependent success probability of oblivious AA on a small compiled circuit as a function of K, and does the amplification failure mode manifest as a K-independent leakage floor?